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 Studierichting: Informatica
 Oefeningenreeks 1B nr. 43.16

$$6(3-i)z^2 + 7 + i = 5(1-i)z$$

$$\Rightarrow (18-6i)z^2 + (-5+5i)z + 7+i = 0$$

$$\Delta = (-5+5i)^2 - 4(18-6i)(7+i)$$

$$= 25 - 50i - 25 - 4(18-6i)(7+i)$$

$$= -50i - 4(126 + 18i - 42i - 6i^2)$$

$$= -50i - 4(132 - 24i)$$

$$= -528 + 46i$$

$$\Delta = (x+yi)^2 = x^2 - y^2 + 2xyi = a + bi = -528 + 46i$$

$$\Rightarrow \begin{cases} x^2 - y^2 &= -528 \\ x^2 + y^2 &= \sqrt{a^2 + b^2} \text{ (de modulus)} \\ 2xy &= 46 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 - y^2 &= -528 \\ x^2 + y^2 &= \sqrt{(-528)^2 + 46^2} \\ 2xy &= 46 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 - y^2 &= -528 \\ x^2 + y^2 &= 530 \\ 2xy &= 46 \end{cases}$$

$$\Rightarrow \begin{cases} -2y^2 &= -528 - 530 \\ 2x^2 &= 530 - 528 \\ 2xy &= 46 \end{cases}$$

$$\Rightarrow \begin{cases} 2y^2 &= 1058 \\ 2x^2 &= 2 \\ 2xy &= 46 \end{cases}$$

$$\Rightarrow \begin{cases} y &= \pm 23 \\ x &= \pm 1 \\ 2xy &= 46 \end{cases}$$

$$xy = 23 > 0 \Rightarrow \sqrt{\Delta} = \pm(1 + 23i)$$

$$z_1 = \frac{(5-5i) + (1+23i)}{2(18-6i)}$$

$$= \frac{6+18i}{36-12i}$$

$$= \frac{1+3i}{6-2i}$$

$$\begin{aligned}
&= \frac{(1+3i)(6+2i)}{(6-2i)(6+2i)} \\
&= \frac{6+2i+18i+6i^2}{36+4} \\
&= \frac{20i}{40} = \frac{i}{2} \\
z_2 &= \frac{(5-5i)-(1+23i)}{2(18-6i)} \\
&= \frac{4-28i}{36-12i} \\
&= \frac{1-7i}{9-3i} \\
&= \frac{(1-7i)(9+3i)}{(9-3i)(9+3i)} \\
&= \frac{9+3i-63i-21i^2}{81+9} \\
&= \frac{30-60i}{90} = \frac{1-2i}{3} = \frac{1}{3} - \frac{2i}{3} \\
z &\in \left\{ \frac{i}{2}, \frac{1}{3} - \frac{2i}{3} \right\}
\end{aligned}$$

References